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Date: 21 / 04 / 2026
Duration: 1 hour 45

CONTINUOUS ASSESSMENT 2
III YEAR – SEMESTER 6
CSE23AE304 - PCP V
(B.Tech. E01,E02,E03,E04)

SET C

Dataset C – Online Course Platform

Each course object contains:

- courseId
- title
- instructor
- category
- price
- rating
- enrolledStudents
- duration
- isActive

Dataset contains **inconsistent and invalid entries**.
Students must handle them appropriately.

Application Requirements

You must build a React application using:

- React Context
- useReducer
- React Router

Mandatory Routes

/courses
/courses/:id
/filter
/stats

Mandatory Rules

- All data must be stored in **Context (Single Source of Truth)**
- All updates must go through **Reducer only**
- No hardcoding dataset
- No storing derived values in state
- Must use **map, filter, reduce**

Testing Requirements (Mandatory for Evaluation)

1. Use test IDs:

data-testid="total-courses"
data-testid="active-courses"
data-testid="inactive-courses"
data-testid="course-item"

2. Expose computed values:

```
window.appState = {  
  totalCourses,  
  activeCourses,  
  inactiveCourses  
};
```

Question 1 – Display Valid Courses

Route: /courses

Requirements:

Display only **valid courses**

Valid Course Conditions:

- title must exist and not empty
- price > 0
- enrolledStudents > 0
- isActive must be boolean

Constraints:

- Must use .filter()
- Must use Context data
- Must not mutate dataset

Edge Cases:

- Missing title → display "Untitled Course"
- Invalid rating → do not display rating

Question 2 – Course Detail View

Route: /courses/:id

Requirements:

Display full course details
Calculate:

revenue = price × enrolledStudents

Constraints:

- Must compute dynamically
- Must validate ID

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Edge Cases:

- Invalid ID → show “Course not found”
- Missing values → handle safely

Question 3 – Filter Courses by Category

Route: /filter

Requirements:

Filter courses by **category**

Constraints:

- Must use .filter()
- Must be case-insensitive
- Must not refetch data

Validation:

- Missing input → show error
- Invalid category → show “No courses found”

Interdependency:

- Must filter only **valid courses (Q1 rules)**

Question 4 – Course Activation Logic

Route: /courses

Requirements:

Add functionality to **toggle course active/inactive**

Business Logic (Buggy Requirement):

- If enrolledStudents ≥ 1000:
 - course must automatically be active

Constraints:

- Must use reducer
- Must not directly mutate state

Edge Cases:

- Invalid course → ignore
- Already active → no duplicate update

Interdependency:

Changes must reflect in:

- Course list
- Stats page

Question 5 – Courses Analytics Dashboard

Route: /stats

Requirements:

Display:

- Total valid courses
- Active courses
- Inactive courses

Constraints:

- Must use .reduce()
- Must not store values in state
- Must compute dynamically

Edge Cases:

- Ignore invalid courses
- Invalid isActive → exclude

Expected Output:

```
{
  "totalCourses": 30,
  "activeCourses": 22,
  "inactiveCourses": 6
}
```

Testing Requirement:

```
<div data-testid="total-courses">30</div>
<div data-testid="active-courses">22</div>
<div data-testid="inactive-courses">6</div>
```

Global State Requirement:

```
window.appState = {
  totalCourses: 30,
  activeCourses: 22,
  inactiveCourses: 6
};
```